

Fault Detection Method for Dust Removal Equipment Based on Digital Twin and OCSVM Optimized by Clonal Selection

1st Ye Tian

Hubei China Tobacco Industry Co., Ltd. Hubei China Tobacco Industry Co., Ltd. Hubei China Tobacco Industry Co., Ltd.
Wuhan, China
sx_tiany@sx.hbtobacco.cn

2nd Hui Chen

sx_chenhui@sx.hbtobacco.cn

3rd Yingfeng Wei

wyfeng@sx.hbtobacco.cn

4th Kai Zhao

Hubei China Tobacco Industry Co., Ltd. Hubei China Tobacco Industry Co., Ltd. Hubei China Tobacco Industry Co., Ltd.
Wuhan, China
sx_zhaokai@sx.hbtobacco.cn

5th Jie Zhu

925263442@qq.com

6th Yuan Xia

414622150@qq.com

7th Guiyuan Wang

Hubei China Tobacco Industry Co., Ltd.
Wuhan, China
sx_wanggy@sx.hbtobacco.cn

8th Qiyi He

Hubei University of Technology
Wuhan, China
20220014@hbut.edu.cn

9th Wen Zhou

Hubei University of Technology
Wuhan, China
zw_mmwh@hbut.edu.cn

10th Han Wang

Hubei University of Technology
Wuhan, China
wanghan@hbut.edu.cn

11th Ao Xu

Hubei University of Technology
Wuhan, China
102211178@hbut.edu.cn

Abstract—Fault detection for dust removal equipment has always been a common anomaly detection scenario. Currently, dust removal equipment fault detection methods face issues such as data imbalance, low accuracy, and insufficient visualization. To address these issues, a fault detection method based on digital twin and OCSVM optimized by clonal selection is proposed. First, the cubic spline interpolation method is used to fill in missing data. Then, to tackle the issue of data imbalance, the One-Class Support Vector Machine (OCSVM) is introduced for anomaly detection. Meanwhile, leveraging the Clonal Selection Algorithm (CSA)'s ability to search for the optimal solution, this algorithm is applied to optimize OCSVM hyperparameters, improving detection accuracy. Finally, to address visualization issues, digital twin technology is used to visualize the dust removal system, the data, and the detection results. Experimental results show that this method effectively monitors anomalies in dust removal equipment, offering higher monitoring accuracy compared to traditional methods and allowing users to intuitively understand the environmental conditions, facilitating timely adjustments.

Index Terms—cubic spline interpolation, digital twin, Clonal Selection Algorithm, One-Class Support Vector Machine, anomaly detection

I. INTRODUCTION

In modern industrial manufacturing, the normal operation of equipment is crucial for both production efficiency and safety. Dust removal equipment anomaly detection, as a common application scenario, not only affects normal operations but may also pose threats to safety and the environment. Efficient anomaly detection methods have long been a critical research topic, and anomaly detection methods based on deep learning and machine learning have garnered widespread attention due to their efficiency and accuracy.

In the field of machine learning, He *et al.* [1] proposed a fault detection method based on the k-nearest neighbor rule (FD-KNN), which can naturally handle possible nonlinearity in the data. Guo *et al.* [2] introduced an improved fault detection method based on kNN, comparing it with MPCA, FD-kNN, and PC-kNN in the semiconductor industry, showing better performance but limited capabilities in handling high-dimensional data. Support vector machines (SVMs), with their powerful classification ability, have demonstrated superiority in fault detection with high-dimensional data. Peng *et al.* [3] proposed a novel fault detection and classification method based on sparse-constrained non-negative matrix factorization

(NMFSC) and structural support vector machines (Structural SVM). Onel *et al.* [4] developed a new feature selection algorithm based on nonlinear SVMs, applied in continuous process monitoring and fault detection. Yang *et al.* [5] combined principal component analysis and recursive feature elimination with SVM for fault detection. However, SVM is sensitive to missing data and its performance is limited when handling large-scale training samples. One-class support vector machine (OCSVM), as a powerful unsupervised anomaly detection algorithm, effectively addresses the issue of data imbalance and has become one of the popular methods in anomaly detection [6]. Martí *et al.* [7] proposed a method combining a fast and high-quality segmentation algorithm with OCSVM for anomaly detection in turbines, effectively addressing the challenge of insufficient labeled data. Pang *et al.* [8] introduced a hybrid algorithm that integrates vector quantization and OCSVM (VQ-OCSVM).

Deep learning technologies have also made significant progress in the field of anomaly detection. Chadha *et al.* [9] compared deep stacked networks with sparse stacked autoencoders for process fault detection, concluding that the sparse stacked autoencoder performed better. Yu *et al.* [10] proposed the stacked denoising autoencoder (SDAE) for process pattern recognition in manufacturing. They later introduced the manifold regularized stacked autoencoder (MRSAE) for fault detection in complex industrial processes [11]. Chen *et al.* [12] proposed a one-dimensional convolutional autoencoder (1D-CAE) for multivariate process fault detection. Li [13] introduced a deep learning method combining a stacked autoencoder (SAE) and long short-term memory (LSTM) neural network for anomaly detection in mechanical equipment, achieving high detection accuracy. Park *et al.* [14] provided a mechanism based on deep learning for rapidly detecting anomalies in multiple devices. Fu *et al.* [15] proposed a re-optimized deep autoencoder (R-DAE) for unsupervised anomaly detection in gas turbines, showing excellent detection performance. Li *et al.* [16] proposed an unsupervised anomaly detection method called Variable Cumulative Error Anomaly Detection (VCEAD) based on autoencoders and temporal convolutional networks, with experimental results showing good performance. Chen *et al.* [17] introduced a wind turbine sequence state monitoring data evaluation method based on LSTM and autoencoders (AE), using LSTM and AE to build a performance evaluation model, and developed an adaptive threshold estimation method based on support vector regression to identify anomalous instances. Ehsani *et al.* [18] proposed an anomaly detection method based on convolutional autoencoders (Conv-AE) for DPMU measurements in power distribution systems, without the need for preprocessing or feature extraction. Li *et al.* [19] introduced an unsupervised anomaly detection method based on memory-augmented recurrent convolutional autoencoders (MARCAE), improving anomaly detection accuracy and noise robustness.

Based on the aforementioned research, anomaly detection primarily faces challenges such as data imbalance, high-dimensional data model construction, difficulty in selecting

model hyperparameters, and insufficient visualization. This study proposes a fault detection method for dust removal equipment based on digital twin and OCSVM optimized by clonal selection to address these issues. While OCSVM has advantages in handling imbalanced data, its detection performance is affected by hyperparameters. The Clonal Selection Algorithm (CSA), with its advantages in global optimization and parallel search, has been widely used in multi-objective optimization problems and can also be applied to optimize OCSVM hyperparameters, compensating for its shortcomings. Experiments show that the proposed detection method demonstrates excellent performance. Additionally, digital twin technology is used to visualize the dust removal equipment and feature data, integrating the detection model into the system, which enhances its practicality.

II. METHODOLOGY

A. Model Framework

The architecture of the dust removal equipment fault detection method based on digital twin and OCSVM optimized by clonal selection includes data acquisition and preprocessing, detection model training and testing, and the construction of the digital twin system. First, feature data from the dust removal equipment are collected, and missing values are filled. During the model training and testing phase, normal data samples are used for training, while the remaining data serve as the test set. The CSA is employed to continuously adjust the OCSVM parameter combinations to obtain the optimal detection model. Finally, digital twin technology is used to visualize the dust removal equipment and data, and the trained detection model is integrated for real-time detection, as shown in Fig. 1.

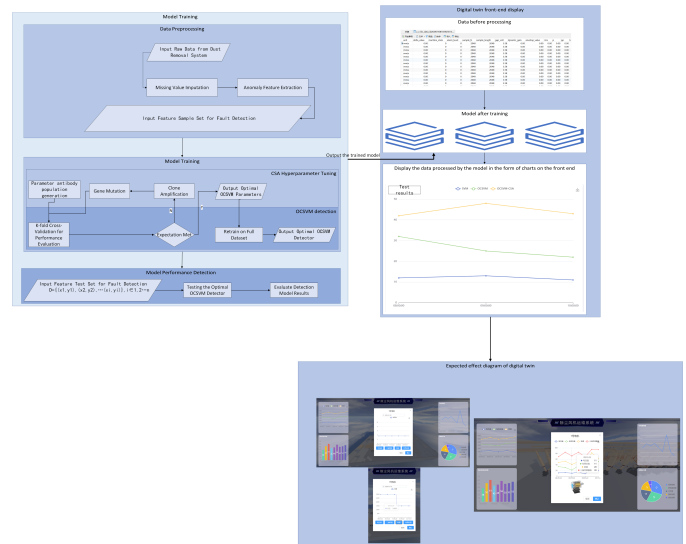


Fig. 1. The Architecture of Dust Removal Equipment Fault Detection Method Based on Digital Twin and OCSVM Optimized by Clonal Selection.

B. Data Preprocessing

The feature data of the environment where the dust removal equipment is located is subjected to missing data imputation, followed by data labeling for normal and abnormal conditions.

Missing data imputation: A cubic spline interpolation method is adopted, which is a method of fitting a smooth continuous curve through a set of discrete points. The formula is as follows:

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3, \quad (1)$$

$$i = 0, 1, 2, \dots, n-1$$

where $S(x)$ is the cubic spline interpolation function containing $n+1$ data points, and $S_i(x)$ is the expression of $S(x)$ in the interval $[x_i, x_{i+1}]$, with the coefficients a_i, b_i, c_i, d_i to be determined. Considering the continuity of $S_i(x)$, its first and second derivatives, and setting the first derivative at the endpoints of the curve to zero, the expression for $S_i(x)$ is:

$$S_i(x) = y_i + \frac{m_i}{2}(x - x_i)^2 + \frac{m_{i+1} - m_i}{6h_i}(x - x_i)^3 \quad (2)$$

where:

$$a_i = \frac{y_{i+1} - y_i}{h_i} - \frac{h_i}{6}(m_{i+1} + 2m_i) \quad (3)$$

Here, $h_i = x_{i+1} - x_i$ is the step size, and m_i is the second derivative at x_i . By solving this, we can obtain the cubic polynomial expression for each segment $S_i(x)$, and then connect them to form the overall spline curve.

The imputed feature data is labeled for subsequent anomaly detection model training and testing.

C. CSA-OCSVM

By leveraging the training set, the CSA continuously optimizes the parameters of the OCSVM, ultimately obtaining the optimal OCSVM detection model. The detailed process is shown in Fig. 2.

The training and detection process of the OCSVM continuously verifies the validity of the parameter combinations provided by the CSA until the model reaches the preset threshold or the algorithm iterates to the maximum number of times. The principle of OCSVM is to find a hyperplane that maximizes the distance from the origin to the boundary of the dataset, and the optimal hyperplane parameters are obtained by solving the following optimization problem:

$$\min_{\omega, \rho, \xi_i} \left(\frac{1}{2} \|\omega\|^2 + \frac{1}{vl} \sum_{i=1}^l \xi_i - \rho \right) \quad (4)$$

subject to:

$$(\omega \cdot \Phi(x_i)) \geq \rho - \xi_i, \quad \xi_i \geq 0 \quad (5)$$

where ξ_i denotes the slack variable, $1/vl$ is the penalty factor, ρ represents the margin from the origin to the hyperplane, x_i is the sample point, Φ is the mapping function, and ω is

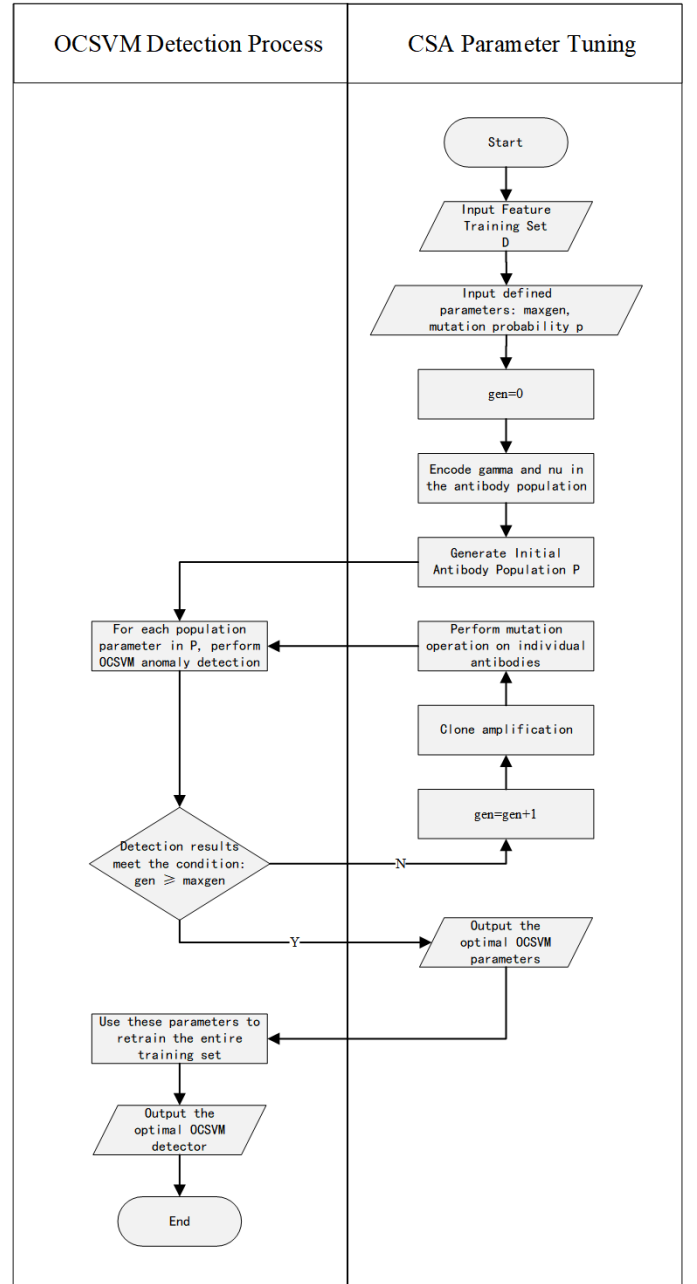


Fig. 2. CSA-OCSVM Process.

the normal vector of the separating hyperplane. The decision function is derived as follows:

$$f(x) = \text{sgn}((\omega \cdot \Phi(x_i)) - \rho) = \text{sgn} \left(\sum_{i=1}^n \alpha_i k(x_i, x) - \rho \right) \quad (6)$$

where α_i are the Lagrange multipliers and $k(x_i, x)$ is the kernel function, which is chosen as the radial basis function. Based on the sign of the function value, the sample is determined to be normal or abnormal.

The CSA parameter tuning process mainly involves setting the maximum number of iterations and the mutation rate.

The former ensures that the process does not fall into an infinite loop and waste resources, while the latter sets the mutation probability to enhance population diversity. This process encodes the kernel function parameter γ and the upper bound of the training error ν . Specifically, the tuning process can be divided into four key steps: population generation, fitness calculation, cloning, and mutation.

Population generation: Encode the parameters γ and ν that need to be optimized and generate N individuals to form a population, that is, multiple parameter combinations.

Fitness calculation: For each individual in the population, OCSVM is used for training and testing, and the detection accuracy is set as the fitness. The higher the fitness, the better the parameter combination. The accuracy is calculated as shown in equation (7).

Cloning: Individuals in the population are cloned, and the number of clones is proportional to the fitness, i.e., the higher the fitness, the more clones are generated.

Mutation: Random mutations are applied to the cloned individuals to increase diversity and explore more parameter combinations. The fitness values are then recalculated, and the top N individuals with the highest fitness are retained as the next generation of the population.

The above steps are repeated until the maximum number of generations is reached, after which the iteration stops, and the optimal OCSVM detection model is obtained.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

D. Digital Twin Visualization

The trained OCSVM anomaly detection model is integrated with digital twin technology to achieve real-time anomaly detection and virtual scenario visualization for dust removal equipment. Digital twin technology involves the virtual mapping of real physical entities by constructing software models to simulate real-world scenarios. The specific steps are as follows.

Modeling and Virtualization: Based on the physical structure and working principles of the components of the dust removal equipment, high-fidelity 3D digital twin models are constructed using computer-aided design (CAD) and simulation software.

Anomaly Detection Module Integration: The OCSVM-based anomaly detection model is integrated into the anomaly detection module of the digital twin system for real-time monitoring of the dust removal equipment's operating status.

Data Collection and Input: Various sensors deployed on-site collect real-time data on the operation of the dust removal equipment and related environmental feature data, transmitting them to the digital twin system as input for the anomaly detection model.

Anomaly Detection and Visualization: The anomaly detection model analyzes the input data and raises an alarm if an anomaly is detected. Meanwhile, the digital twin system renders the real-time data and detection results on the user interface, allowing the operator to intuitively view the current

operating status of the dust removal equipment and any detected anomalies. Fig.3 illustrates the visualization of motor 1 data for the dust removal equipment in the digital twin system.



Fig. 3. Visualization of Dust Removal Equipment Motor No. 1 Data

III. EXPERIMENTS AND ANALYSIS

A. Dataset

The dataset used in the experiment is collected from sensors related to the dust removal equipment. Various feature data are collected and undergo a unified time-series merging and labeling process, marking normal and abnormal data for the training and testing of the anomaly detection model. Additionally, missing values in the collected data are imputed using cubic spline interpolation before the labeling process, ensuring data completeness for model training.

B. Experimental Design and Analysis

The experiment designed a comparison of three methods in terms of accuracy, recall, precision, and F1 score: Isolation Forest (IForest), OCSVM, and the proposed CSA-OCSVM. The experimental coding environment is Python 3.9. The experimental comparison results are shown in Table I and Fig.4. Table I shows the comparison of various metrics, and the specific metric formulas are described in equations (8-10). Fig.4 shows the ROC curves for each method, where Tuned OCSVM represents the CSA-OCSVM method. The experimental results demonstrate that the OCSVM, optimized by the clonal selection algorithm, exhibits excellent detection performance, outperforming the other two methods across all metrics.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

TABLE I
ANOMALY DETECTION RESULTS OF VARIOUS METHODS

Model	Accuracy	F1-score	Precision	Recall	AUC
IForest	64%	23%	15%	56%	0.61
OCSVM	64%	35%	21%	100%	0.80
CSA-OCSVM	99%	94%	89%	100%	0.99

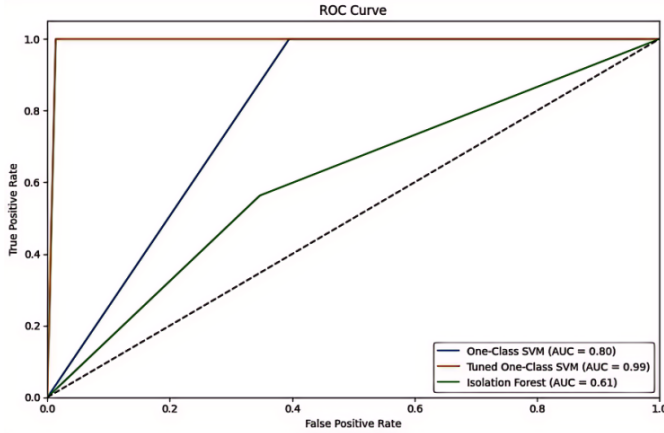


Fig. 4. ROC Curves of Various Methods

IV. CONCLUSION

This paper develops a detection model based on a Clonal Selection Algorithm optimized One-Class Support Vector Machine and achieves promising detection results. By integrating the trained model into the digital twin system, the visualization and real-time monitoring of the dust removal equipment are realized, effectively enhancing its usability. Future work will focus on further optimizing the detection model, such as incorporating feature selection algorithms, to further improve its detection performance.

ACKNOWLEDGMENT

This work was supported in part by NSFC (<http://www.nsf.gov.cn/>) under Grant 62202147, Grant 62376089, and Grant 42201464; in part by the Wuhan Science and Technology Bureau 2022 Knowledge Innovation Dawning Plan Project under Grant 2022010801020270; and in part by the Scientific Research Project of Natural Resources Department of Hubei Province under Grant ZRZY2023KJ13.

REFERENCES

- [1] Q. P. He and J. Wang, "Fault detection using the k-nearest neighbor rule for semiconductor manufacturing processes," *IEEE Transactions on Semiconductor Manufacturing*, vol. 20, no. 4, pp. 345–354, 2007.
- [2] X. P. Guo, J. Yuan, and Y. Li, "KPCS-kNN based fault detection for batch processes," in *2013 International Conference on Machine Learning and Cybernetics*, vol. 2, IEEE, pp. 698–703, 2013.
- [3] Y. Peng, X. Chen, Q. Ye, *et al.*, "Fault detection and classification in chemical processes using NMFSC and structural SVMs," *The Canadian Journal of Chemical Engineering*, vol. 92, no. 6, pp. 1016–1023, 2014.
- [4] M. Onel, C. A. Kieslich, Y. A. Guzman, *et al.*, "Simultaneous fault detection and identification in continuous processes via nonlinear support vector machine based feature selection," in *Computer Aided Chemical Engineering*, Elsevier, vol. 44, pp. 2077–2082, 2018.

- [5] C. Yang and J. Hou, "Fed-batch fermentation penicillin process fault diagnosis and detection based on support vector machine," *Neurocomputing*, vol. 190, pp. 117–123, 2016.
- [6] M. A. Panza, M. Pota, and M. Esposito, "Anomaly detection methods for industrial applications: A comparative study," *Electronics*, vol. 12, no. 18, pp. 3971, 2023.
- [7] L. Martí, N. Sanchez-Pi, J. M. M. López, *et al.*, "On the combination of support vector machines and segmentation algorithms for anomaly detection: A petroleum industry comparative study," *Journal of Applied Logic*, vol. 24, pp. 71–84, 2017.
- [8] J. Pang, X. Pu, and C. Li, "A hybrid algorithm incorporating vector quantization and one-class support vector machine for industrial anomaly detection," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 12, pp. 8786–8796, 2022.
- [9] G. S. Chadha and A. Schwung, "Comparison of deep neural network architectures for fault detection in Tennessee Eastman process," in *2017 22nd IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, IEEE, pp. 1–8, 2017.
- [10] J. Yu, X. Zheng, and S. Wang, "A deep autoencoder feature learning method for process pattern recognition," *Journal of Process Control*, vol. 79, pp. 1–15, 2019.
- [11] J. Yu and C. Zhang, "Manifold regularized stacked autoencoders-based feature learning for fault detection in industrial processes," *Journal of Process Control*, vol. 92, pp. 119–136, 2020.
- [12] S. Chen, J. Yu, and S. Wang, "One-dimensional convolutional auto-encoder-based feature learning for fault diagnosis of multivariate processes," *Journal of Process Control*, vol. 87, pp. 54–67, 2020.
- [13] Z. Li, J. Li, Y. Wang, *et al.*, "A deep learning approach for anomaly detection based on SAE and LSTM in mechanical equipment," *The International Journal of Advanced Manufacturing Technology*, vol. 103, pp. 499–510, 2019.
- [14] H. Park, D. Park, and S. Kim, "Anomaly detection of operating equipment in livestock farms using deep learning techniques," *Electronics*, vol. 10, no. 16, pp. 1958, 2021.
- [15] S. Fu, S. Zhong, L. Lin, *et al.*, "A re-optimized deep auto-encoder for gas turbine unsupervised anomaly detection," *Engineering Applications of Artificial Intelligence*, vol. 101, pp. 104199, 2021.
- [16] Z. Li, Y. Sun, L. Yang, *et al.*, "Unsupervised machine anomaly detection using autoencoder and temporal convolutional network," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–13, 2022.
- [17] H. Chen, H. Liu, X. Chu, *et al.*, "Anomaly detection and critical SCADA parameters identification for wind turbines based on LSTM-AE neural network," *Renewable Energy*, vol. 172, pp. 829–840, 2021.
- [18] N. Ehsani, F. Aminifar, and H. Mohsenian-Rad, "Convolutional autoencoder anomaly detection and classification based on distribution PMU measurements," *IET Generation, Transmission & Distribution*, vol. 16, no. 14, pp. 2816–2828, 2022.
- [19] W. Li, Z. Shang, J. Zhang, *et al.*, "A novel unsupervised anomaly detection method for rotating machinery based on memory augmented temporal convolutional autoencoder," *Engineering Applications of Artificial Intelligence*, vol. 123, pp. 106312, 2023.